

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education KCSE PREDICTION SERIES 1



233/3

Chemistry (Practical)

Paper 3

Time: 2¼ Hours

CONFIDENTIAL

In addition to the apparatus and fittings available in the chemistry laboratory, each candidate should have:

- About 100cm³ of solution A
- About 100cm³ of solution B
- About 60cm³ of solution C
- One burette
- One pipette
- One thermometer
- 0.5g of solid E
- 0.5g of solid F
- Blue and red litmus papers
- 0.2g of solid sodium hydrogen carbonate/sodium carbonate.
- Two conical flasks
- Five clean dry test tubes
- Test tube holder
- One 100 ml plastic beaker
- 10 ml measuring cylinder
- 50 ml measuring cylinder.
- 1 boiling tube.
- Universal indicator paper.
- PH chart.
- 250 cm³ plastic beaker.
- Source of heat
- Phenolphthalein indicator.
- Bromine water supplied with a dropper
- Acidified potassium manganate (VII) supplied with a dropper
- 2M sodium hydroxide supplied with a dropper
- 2 M sulphuric (VI) acid supplied with a dropper
- 2M lead (II) nitrate supplied with a dropper

Note:

- Solution A is 1 M sodium hydroxide
- Solution B is 0.128M hydrochloric acid
- Solution C is 1 M ethanoic acid
- Solid E is calcium chloride
- Solid F is oxalic acid

Access to:

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Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- a) Write your name and index number in the spaces provided above.
- b) Mathematical tables and electronic calculators may be used for calculations
- c) *All* working must be clearly shown where necessary
- d) Answer **all** the questions in the **spaces provided** in the question paper

FOR EXAMINER'S ONLY

QUESTION	MAXIMUM SCORE	CANDIDATES SCORE
1	22	
2	08	
3	10	

TOTAL	40	
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1. You are provided with:

- Sodium hydroxide labelled solution **A**
- 0.128M hydrochloric acid labelled solution **B**
- Carboxylic acid labelled solution **C**

You are required to:

- a) Standardize solution **D** with solution **B**
- b) Determine the:
 - i. Reaction ratio between sodium hydroxide, solution A and the carboxylic acid, solution **C**
 - ii. Concentration of solution **C** in moles per litre.

Procedure1

Pipette 25 cm³ of solution A into a 250 cm³ beaker. Add distilled water to make up to 150 cm³. Label this solution D.

Fill the burette with solution B. Pipette 25cm³ of solution **D** into a 250cm³ conical flask. Add 2 dropsofphenolphthalein indicator and titrate with solution **B**. Record your results in table 1. Repeat the titration two more times and complete the table

Table 1

	I.	II.	III.
Final burette reading			
Initial burette reading			
Volume of solution B used (cm ³)			

(4 marks)

a) Determine the average volume of solution **B** used.

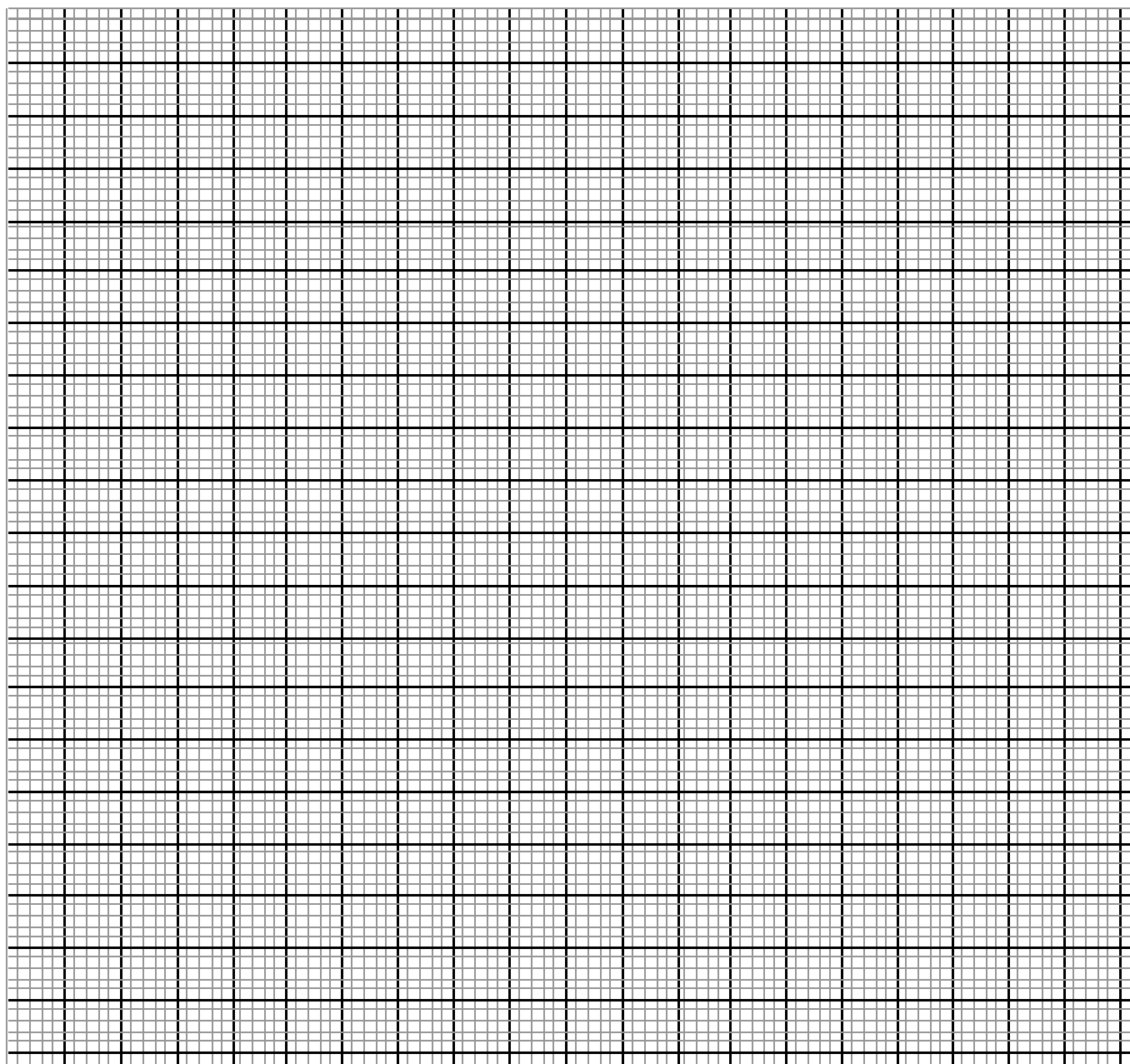
(1mark)

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- a) On the grid provided, plot a graph of ΔT (vertical axis) against the volume of sodium hydroxide, solution A (3marks)



- b) From the graph, determine the volume of sodium hydroxide, solution A required to neutralize the carboxylic acid (1mark)

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c) Calculate the volume of carboxylic acid, solution C used for neutralization (1mark)

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d) Calculate the:

i. Ratio between the volumes of solutions A and C (1mark)

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ii. Concentration in moles per litre of the carboxylic acid, solution C.

(Assume that the volume ratio is the same as the mole ratio) (2marks)

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2. You are provided with solid E. Carry out the tests below and record your observations and inferences in the spaces provided.

Place all the solid E in a boiling tube, add 10cm³ of distilled water and shake.

OBSERVATION	INFERENCE

(1mark)

(1mark)

- I. To about 1cm³ of the solution, add 2M sodium hydroxide drop wise until in excess

OBSERVATION	INFERENCE

(1mark)

(1mark)

- II. Place 1cm³ of the solution in a test-tube and add 2 to3 drops of 2M sulphuric (VI) acid.

OBSERVATION	INFERENCE

(1mark)

(1mark)

- III. To about 1cm³ of the solution, add 4-5 drops of 2M lead (II) nitrate solution and warm.

OBSERVATION	INFERENCE

(1mark)

(1mark)

3. You are provided with solid F. Carry out the tests below and record your observations and inferences in the spaces provided.

Place all of solid F into a boiling tube. Add 10cm³ of distilled water and shake well. Use 2cm³ portions of the mixture for the following reactions.

OBSERVATION	INFERENCE

(1mk)

(1mk)

- a) Test the PH of first portion using the universal indicator paper.

OBSERVATION	INFERENCE

(1mark)

(1mark)

b) To the second portion, add three drops of bromine water.

OBSERVATION	INFERENCE

(1mark)

(1mark)

c) To the third portion, add 2 drops of acidified potassium manganate (VII) and shake well

OBSERVATION	INFERENCE

(1mark)

(1mark)

d) To the fourth portion, add all the sodium hydrogen carbonate provided.

OBSERVATION	INFERENCE

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(1mark)

(1mark)

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